

MODULE 1: PLATFORM FOUNDATIONS

Interactive 3-Hour Rhythm

Session 1.1: Course Orientation & The Fundamentals of Platform Businesses

Linear value chains vs. two-sided ecosystem orchestration, transaction cost economics, and the asset-light growth model.

PART 1 (LECTURE)	PART 2 (LAB PRACTICE)	PART 3 (EXECUTIVE MEMO)	CORE METHOD
60 Minutes	60 Minutes	60 Minutes	SEC 10-K Reconciliation

Learning Objectives & Core Vocabulary

Clear definitions for students with zero prior platform economics background.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Deconstruct Value Models:** Distinguish linear pipeline production from multi-sided ecosystem orchestration.
- **Apply Economic Theory:** Explain how digital software protocols eliminate Coasean transaction friction.
- **Audit Financial Statements:** Calculate fixed asset intensity and labor leverage from SEC 10-K regulatory filings.
- **Formulate Strategy:** Advise executive management on navigating legacy-to-platform channel conflict.

FOUNDATIONAL TERMS

Vocabulary

Five Essential Concepts

- **Pipeline Business:** Linear step-by-step value chain where the firm owns inventory and controls inputs to outputs.
- **Platform Business:** Digital infrastructure that matches external producers and consumers to enable direct exchanges.
- **Intermediation:** Automated coordination of transactions without taking balance-sheet ownership of physical assets.
- **Asset-Light Model:** Scaling revenue rapidly by substituting software code for capital expenditure and physical property.
- **Operational Leverage:** Growing transactions exponentially while keeping corporate headcount and fixed overhead flat.

Theory 1: Linear Pipelines vs. Platform Ecosystems

Van Alstyne, Parker, and Choudary (2016) - "Pipelines, Platforms, and the New Rules of Strategy"

Strategic Dimension	Linear Pipeline Model	Multi-Sided Platform Model
Value Architecture	Linear supply chain: Supplier → Manufacturer → Distributor → Consumer.	Ecosystem matrix: Orchestrates external exchanges between producers and consumers.
Asset Base & PP&E	Heavy property, plant, and equipment (PP&E) owned directly on balance sheet.	Asset-light software infrastructure; physical assets owned by external ecosystem participants.
Scale Mechanism	Supply economies of scale: Lowering per-unit fixed costs through physical plant volume.	Demand economies of scale: Value accelerates via network effects as participant count grows.
Growth Bottleneck	Capital expenditure (CapEx), physical construction lead times, and factory capacity.	Algorithmic match quality, user trust, liquidity density, and transaction safety.

STRATEGIC TAKEAWAY FOR BUSINESS LEADERS
Pipelines compete by optimizing internal processes and controlling proprietary resources. Platforms compete by governing external interactions and orchestrating vibrant community ecosystems.

Theory 2: Transaction Cost Economics & Digital Networks

Ronald Coase (1937) "The Nature of the Firm" & Oliver Williamson (1981) "The Economics of Organization"

FRICTION 1

Discovery

Search Costs

The time, labor, and money required for buyers and sellers to discover compatible trading partners in an open market.

- **Legacy World:** Travel agencies, printed catalogs, local physical directories.
- **Digital Platform:** Algorithmic real-time indexing, search filters, instant price discovery.

FRICTION 2

Agreement

Contracting Costs

The legal, negotiation, and escrow expenses necessary to reach binding agreements and protect counterparty interests.

- **Legacy World:** Individual bespoke contracts, lawyers, notarized guarantees.
- **Digital Platform:** Standardized terms of service, programmatic insurance policies, integrated escrow.

FRICTION 3

Fulfillment

Coordination Costs

The operational overhead required to enforce terms, verify performance, and resolve bilateral transaction disputes.

- **Legacy World:** On-site human inspection, customer service hotlines, court arbitration.
- **Digital Platform:** Two-way verified reputation systems, automated refund protocols, telemetry tracking.

THE COASEAN DIGITAL SHIFT

When digital software protocols reduce search, contracting, and coordination costs toward zero, transactions that previously required massive integrated corporations can now occur efficiently across open decentralized networks.

Accessible Micro-Examples Across Industries

Translating abstract theory into everyday corporate and consumer realities.

HOSPITALITY

Accommodation

Hotel vs. Marketplace

- **Marriott (Pipeline):** Purchases prime urban land, constructs concrete hotels, hires cleaning staff, and sells room-nights.
- **Airbnb (Platform):** Builds cloud software connecting homeowners with global travelers without owning a single bedroom.
- **Key Difference:** Marriott must spend billions to add 10,000 rooms; Airbnb adds 10,000 listings with lines of code.

PUBLISHING

Media

Publisher vs. KDP

- **Traditional Publisher (Pipeline):** Employs editors to reject 99% of manuscripts, prints paper books, warehouses pallets.
- **Amazon KDP (Platform):** Provides automated web portal where any author uploads a file and sells directly worldwide.
- **Key Difference:** Legacy editorial gatekeepers replaced by community customer reviews and algorithmic ranking.

FINANCIAL SERVICES

Fintech

Retail Bank vs. Stripe

- **Commercial Bank (Pipeline):** Builds physical marble branches, manages proprietary teller networks, sells manual wire transfers.
- **Stripe (Platform):** Publishes seven lines of Javascript code allowing developers to accept payments globally in minutes.
- **Key Difference:** Physical distribution replaced by developer API integration and cloud clearing protocols.

Case Dissection: Airbnb vs. Marriott International

Empirical confrontation between physical asset ownership and software orchestration (2015-2023).

HISTORICAL TRAJECTORY

2015-2023 Timeline

Strategic Divergence

- Marriott Physical Expansion:** Invested over \$13 billion in cash and debt to acquire Starwood Hotels in 2016, absorbing massive ongoing hotel property leases and renovations.
- Airbnb Software Expansion:** Recruited independent hosts globally via word of mouth and algorithmic search optimization, reaching 7 million active listings by 2023.
- Capital Expenditure Gap:** Marriott allocates hundreds of millions annually to property maintenance; Airbnb operates as an ultra-high cash-flow generator.

OPERATIONAL METRICS

Empirical Comparison

The Power of Software Leverage

Metric (FY2023)	Marriott Int.	Airbnb Inc.
Total Active Listings	~1.6 Million Rooms	Over 7.7 Million Listings
Net PP&E / Assets	~7.4% (\$1,861M)	<0.8% (\$154M)
Corporate Headcount	~411,000 (Managed)	6,889 Employees
Revenue / Employee	~\$57,700 / Worker	~\$1,438,000 / Worker

Lab Transition: 120-Minute Hands-On Execution

Shifting from lecture observation to audited data manipulation and executive decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Audited SEC 10-K Financial Lab

Step-by-step spreadsheet model to compute and verify asset intensity metrics:

- **Dataset:** Open `data/session_1_1_asset_intensity.csv` in Excel or Google Sheets.
- **Step 3 (PP&E Ratio):** In cell `J2`, enter formula `=D2/C2*100` (Net PP&E to Assets).
- **Step 4 (Labor Productivity):** In cell `K2`, enter formula `=(E2*1000000)/G2` (Revenue per Worker).
- **Audited Finding:** Quantify the 25x labor leverage advantage of digital platform orchestration.

EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Dilemma & Board Strategy Memo

Resolve Marriott's real-world corporate challenge with "Homes & Villas":

- **The Dilemma:** Traditional pipeline hotel operators face high CapEx; can Marriott operate a marketplace without cannibalizing franchisees?
- **Key Tradeoffs:** Premium brand quality control vs. rapid host onboarding; franchise owner backlash.
- **Student Deliverable:** Draft a structured 3-bullet decision brief for the Board of Directors: Build internally or acquire an existing rental platform?